MARIE MCGREGOR

Formerly Marie Grace

Boulder, CO

404-989-3108

mariemcgregor05@gmail.com

mgrace31.github.io

PROFILE

I am a 2nd-year PhD student in Computer Science, who brings a robust foundation in computer science to language tasks. I have experience in human-AI teaming and error analysis for NLP and have focused recently on development of explainable metrics for structured meaning representations and integrating them into LLMs. I am committed to finding solutions to human-language centered problems by creating reliable and explainable models.

EDUCATION

University of Colorado Boulder | 2027
Ph.D. Student in Computer Science

University of Colorado Boulder | 2023
Master of Science in Computational Linguistics

Georgia Institute of Technology | 2021
Bachelor of Science in Computer Science, French Minor

SKILLS AND CONCEPTS

Natural language Processing, Data Center Scale Computing, Neuro-Symbolic Approaches to NLP Machine Learning, Artificial Intelligence, Computational Models of Discourse, Algorithms, Linguistics, Database Systems, Probability and Statistics, Computer Organization and Programming, Combinatorics,
Linear Algebra

EXPERIENCE

May 2024-August 2025
NLP Research Intern| Leidos
Created in-depth literature review on causes of LLM hallucinations and mitigation strategies. Explored detection of LLMs using clustering of model hidden states. Conducted experimentation and evaluation of author attribution and personality detection research using logistic regression.

May 2024-August 2024
Online Curriculum Assistant | CU Boulder
Created assignments for the error analysis and evaluation course of CU Boulder's online NLP Master's degree covering binary and multi-class classification, model comparisons, generative model evaluation, and model interpretability.

August 2023- Present
Teaching Assistant | Intro to Computational Thinking | CU Boulder
Instructing multiple sections, creating python course materials, and grading.

May 2021-May 2022
Research and Development Intern | Autonomy Team| Sandia National Labs
Used machine learning to predict a vehicle's intended terminal location, evaluated various model types, conducted hyperparameter optimization, determined training data requirements, and assessed computational burdens during training and testing.

RESEARCH PROJECTS

August 2023- Current
Graduate Researcher | NSF Uniform Meaning Representations (UMR)
Managing human annotations of UMRs for creating cross-linguistically valid meaning representation of natural language. Overseeing automatic conversion of prior schemas to UMR schema. Investigating best evaluation metrics for UMR graphs.

May 2022-May 2023
Graduate Researcher | NSF National Institute for Student-AI Teaming (iSAT)
Designed and implemented an annotation scheme for students' classroom speech to evaluate key linguistic features that are predictive of positive collaboration behavior.

May 2022-August 2022
Graduate Researcher | Offensive Language Error Analysis (OLEA)
Designed and developed a tool for evaluating model performance on offensive and hateful language classification and researched current model performance and limitations to provide meaningful linguistic insights.

May 2019- May 2021
Undergraduate Researcher | Dynamic Adaptive Robotic Technologies (DART)
Used Dynamic Time Warping and Hidden Markov Models in recognition algorithms to identify flightpaths in a flight simulator and implemented control arbitration to create a shared control system between the pilot and an autonomous agent that leads to smoother and more accurate flying.

PUBLICATIONS

https://scholar.google.com/citations?user=vaZD4rsAAAAJ&hl=en